

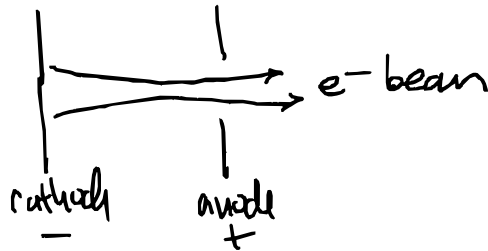
Longitudinal Motion + Phase Stability

Ref. Weidemann Ch. 9, Lee Ch. 3

Forces in direction of motion

→ long. fields or acc. fields

Simple DC accelerator:



→ limit 10's MV

⇒ This leads us to RF acceleration

Note: free EM wave cannot accelerate.

→ No long. field component

→ requires some physical environment/
Boundary Conditions to change the
EM wave into an accelerating field.

Assume we have an oscillating E field:

$$\vec{E}(z, t) = \vec{E}_0 e^{i(\omega t - kz)} = \vec{E}_0 e^{i\phi}$$

↑
phase

phase $\omega t - kz \Rightarrow$ fields vary w. time
 ↑
 ω_{rf} fields vary w. z

Rate of change of momentum:

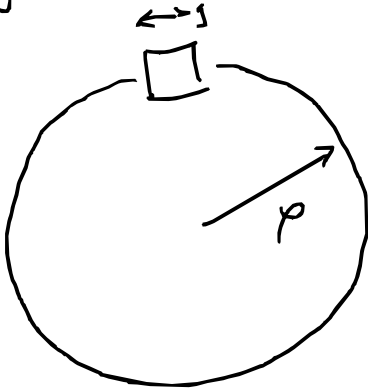
$$\frac{d\vec{p}}{dt} = e \vec{E}(\phi) = \frac{d}{dt}(\gamma m c \beta)$$

Change in kinetic Energy:

$$\Delta E = (\gamma - \gamma_0) m c^2 = e \int E(\phi) dz$$

$\rightarrow$ integrate field across the cavity

Imagine an RF cavity in a ring:



$$g \ll 2\pi\rho$$

$$\tau = \frac{L}{v} = \frac{L}{\beta c}$$

$$\vec{E} = \vec{E}_0 e^{i\phi}$$

For systematic acceleration, the phase of the cavity must reach the same phase each time the particle arrives.

$$\phi_s = \omega t - k z = \text{const.}$$

"synchronous phase"

$$\rightarrow \dot{\phi}_s = \omega_{rf} - k \dot{z} = 0 \quad (\dot{z} = \beta c)$$

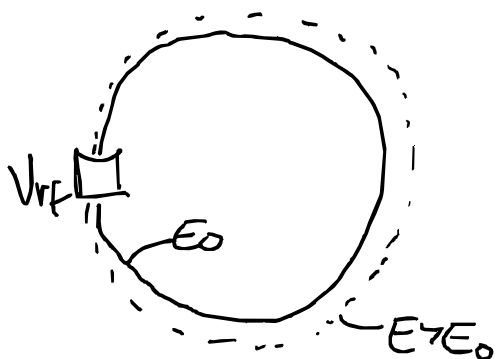
We can meet this condition for $k = \frac{2\pi}{L}$

$$\Rightarrow \omega_{rf} = k \beta c = \frac{2\pi}{L} \beta c = \frac{2\pi}{\tau}$$

$$\omega_{rf} = h \frac{2\pi}{\tau}$$

³
h = harmonic #

Imagine a ring, w. 1 RF cavity and 1 long magnet



How does the revolution freq change w. energy?
 $E \uparrow \rightarrow$ faster?
 $E \uparrow \rightarrow$ longer distance?

The revolution frequency

$$\omega = \frac{2\pi}{\tau} = \frac{2\pi\beta c}{L}$$

Differentiate $\ln(\omega)$

$$\rightarrow \frac{d\omega}{\omega} = -\frac{d\tau}{\tau} = \frac{d\beta}{\beta} - \frac{dL}{L}$$

$$\Rightarrow \frac{d\tau}{\tau} = \frac{dL}{L} - \frac{d\beta}{\beta} \hookrightarrow \frac{d\beta}{\beta} = \frac{1}{\gamma^2} \frac{dP}{P}$$

(comes from relativity formulas)

$$\frac{d\tau}{\tau} = \frac{dL}{L} - \frac{1}{\gamma^2} \frac{\Delta P}{P} \delta$$

$$\left(\delta = \frac{\Delta P}{P} \right)$$

$$= \frac{dL}{L} - \frac{1}{\gamma^2} \delta$$

$$= - \left(\frac{1}{\gamma^2} - \frac{\delta L/L}{\delta} \right) \delta$$

$$= - \left(\frac{1}{\gamma^2} - \alpha_c \right) \delta$$

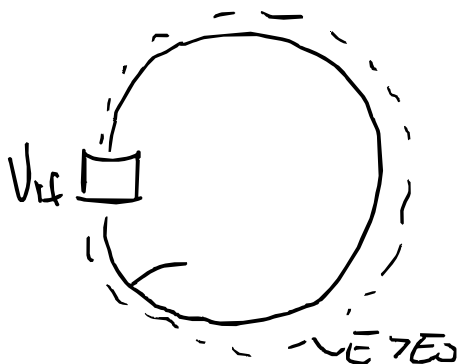
↳ momentum
compaction
factor.

$$\Rightarrow \frac{\delta\tau}{\tau} = -\eta_{tr} \delta$$

↳ when $\eta \rightarrow 0$

$\equiv$ transition energy

What is the momentum compaction factor?



$$\alpha_c = \frac{\delta L / L}{\delta}$$

→ Compares the rate of change of the path length to the rate of change of momentum.

Then the phase slip factor $\gamma_{tr} = \frac{1}{\gamma^2} - \alpha_c$

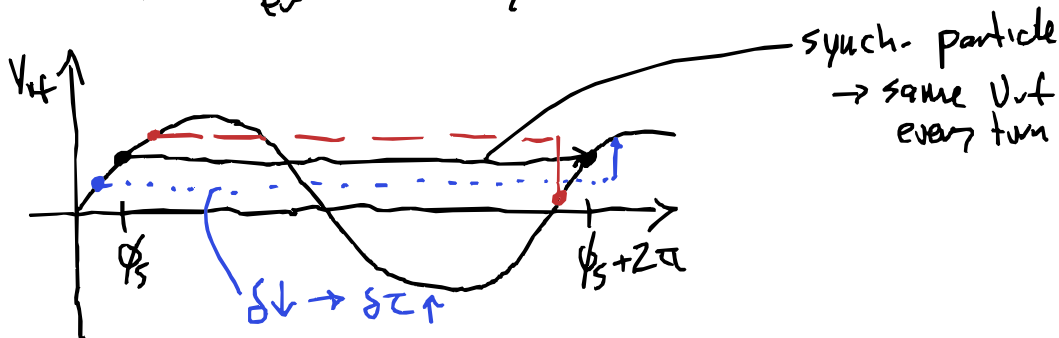
tells you when $\frac{\delta \tau}{\tau}$ changes sign.

Remember: $\delta \tau / \tau$ is the rate of change in the arrival time of the particle in the RF cavity

So, when $\gamma \rightarrow 0 \equiv$ transition energy γ_{tr}

Stability BELOW transition Energy:

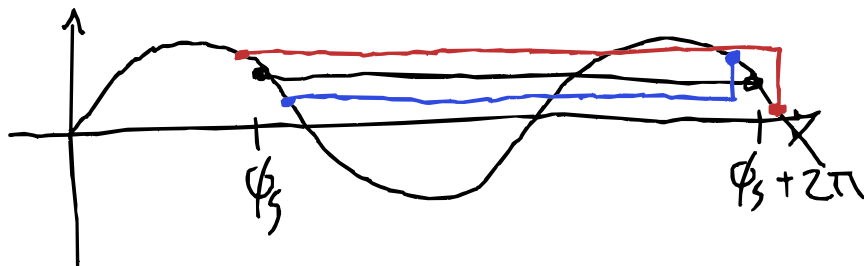
$$\gamma < \gamma_{tr} \rightarrow \eta_{tr} > 0$$



since $\frac{\delta \tau}{\tau} = -\eta_{tr} \delta$

$\uparrow +$, so lower $E \Rightarrow$ longer period.

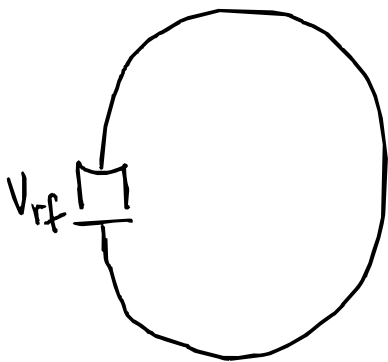
ABOVE transition:



$\rightarrow$ lower E particle has shorter period.

Note: the synchronous phase must be selected based on γ vs γ_{tr}

Longitudinal Eqs of Motion



$$\omega_{rf} = h \omega_s$$

↑
harmonic #

$$\omega_s = \frac{2\pi \beta c}{L}$$

Note: when $\beta \neq 1 \rightarrow$ as $E \uparrow$, $\beta \uparrow \rightarrow \omega \uparrow$

$\beta \sim 1 \rightarrow$ as $E \uparrow$, $\beta \sim 1 \rightarrow L \uparrow + \omega \downarrow$

$$V = \int E(z, t) dz = V \sin(\omega_{rf} t + \phi_s)$$

$\therefore$ The energy gain of the synchronous particle (per revolution) is:

$$\Delta U_s = q V \sin \phi_s$$

For a particle arriving with a time offset δt

$$\rightarrow \Delta U = q V \sin(\omega_{rf} \delta t + \phi_s) \text{ per } \underline{\underline{\text{turn}}}$$

Now, let's define some generic offsets from the synchronous values:

$$U = U_s + \delta U \quad \text{total } E$$

$$p = p_s + \delta p$$

$$\omega = \omega_s + \delta \omega$$

$$\tau = \tau_s + \delta \tau \quad \text{rev. period}$$

$$\varphi = \delta \phi = \phi - \phi_s \quad \text{relative phase.}$$

→ Energy gain per turn:

$$\delta U_s = qV \sin \phi_s$$

$$\delta U = qV \sin \phi = qV \sin(\phi_s + \delta \phi)$$

Energy diff. on the n^{th} turn: (b/w generic particle + sync.)

$$(\delta U)_n = U - U_s$$

At the $(n+1)^{\text{th}}$ turn:

$$(\delta U)_{n+1} = (U + \Delta U) - (U_s + \Delta U_s)$$

↑
prior energy

↑
energy gain per turn

↑
same for
sync. particle

Relative change in energy per turn is then:

$$\begin{aligned}\Delta(\delta U) &= (\delta U)_{n+1} - (\delta U)_n \\ &= [(U + \Delta U) - (U_s + \Delta U_s)] - (U - U_s) \\ &= \Delta U - \Delta U_s \\ &= gV(\sin \phi - \sin \phi_s)\end{aligned}$$

Turn this into a diff. eq'n:

$$\begin{aligned}\frac{d(\delta U)}{dt} &\approx \frac{\Delta(\delta U)}{T_s} \rightarrow \text{Rel. energy gain per turn} \\ &\quad \rightarrow \text{Time per turn.} \\ &= \frac{gV}{2\pi} \omega_s (\sin \phi - \sin \phi_s)\end{aligned}$$

Now, it's useful to define a new canonical energy variable

$$(\delta E, -\delta U) \rightarrow (\omega_{rf} \delta E, W)$$

$$\text{where: } W = -\frac{\delta U}{\omega_{rf}} = -\frac{(U - U_s)}{\omega_{rf}}$$

(Note: preserves phase space area)

Let's look at phases again:

The change in rel. phase is:

$$\Delta\varphi \approx \frac{d\varphi}{dt} \tau_s = \omega_{rf} \delta t$$

After one revolution, the difference in arrival times becomes

$$\Delta(\delta t) = \tau - \tau_s = d\tau = -\eta_{tr} \tau \frac{dp}{p}$$

remember:
phase slip factor

$$\frac{d\tau}{\tau} = -\eta_{tr} \delta$$

Combine these last two eq'n's:

$$\frac{d\varphi}{dt} \approx \frac{\Delta\varphi}{\tau_s} = \frac{\omega_{rf}}{\tau_s} \Delta(\delta t)$$

$$= \frac{\omega_{rf}}{\tau_s} \left(-\eta_{tr} \tau \frac{dp}{p} \right)$$

From relativity, we had:

$$U^2 = p^2 c^2 + m^2 c^4$$

Differentiate:

$$2U \Delta U = 2p c^2 \Delta p$$

$$\swarrow \left(\beta = \frac{pc}{E} \right)$$

$$\rightarrow \frac{\Delta p}{p} = \frac{\Delta U U}{p^2 c^2} = \frac{\Delta U}{U} \frac{U^2}{p^2 c^2} = \frac{1}{\beta^2} \frac{\Delta U}{U}$$

Remember - we defined $W = -\frac{\Delta U}{\omega_{rf}}$

$$\rightarrow \frac{\Delta p}{p} = \frac{1}{\beta^2 U_s} (-\omega_{rf} W)$$

Putting these together now

$$\frac{d\phi}{dt} = -\frac{\omega_{rf}}{c_s} \eta_{tr} \frac{1}{\beta^2 U_s} (-\omega_{rf} W)$$

$$\dot{\phi} = \frac{\omega_{rf}^2 \eta_{tr}}{\beta^2 U_s} W$$

Second derivative (remember $\rightarrow$ we want an Eq'n of motion)

$$\begin{aligned}\frac{d}{dt} \dot{\phi} &= \frac{\omega_{rf}^2 \eta_{tr}}{\beta^2 U_s} \frac{d\omega}{dt} \\ &= \frac{\omega_{rf}^2 \eta_{tr}}{\beta^2 U_s} \frac{qV}{2\pi} (\sin \phi - \sin \phi_s)\end{aligned}$$

$\Rightarrow$ This is our eq'n of motion, relative to the synchronous particle.

$$\ddot{\phi} + \frac{\omega_{rf}^2 \eta_{tr} qV}{2\pi \beta^2 U_s} (\sin \phi_s - \sin \phi) = 0$$

For small oscillations:

$$\begin{aligned}\sin \phi &= \sin(\phi_s + \phi) \\ &\approx \phi \cos \phi_s + \sin \phi_s\end{aligned}$$

$$\begin{aligned}\Rightarrow \ddot{\phi} + \underbrace{\left(\frac{\omega_{rf}^2 \eta_{tr} qV}{2\pi \beta^2 U_s} \cos \phi_s \right)}_{\omega_z^2} \phi &= 0 \\ \ddot{\phi} + \omega_z^2 \phi &= 0\end{aligned}$$

Ω = synchrotron angular frequency

$$\ddot{\phi} + \Omega_s \phi = 0$$

→ Eq'n of motion for oscillatory motion

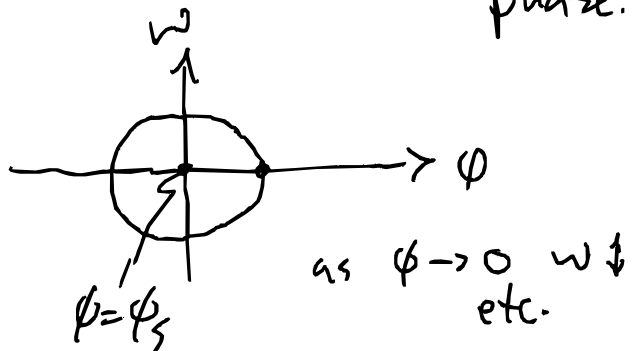
For real values of the sync. freq:

$$\phi = \phi_0 \cos(\Omega t + \chi_i)$$

since $\frac{d\phi}{dt} \propto \omega$ $\left(\frac{d\phi}{dt} \sim \frac{\delta p}{p} \right)$

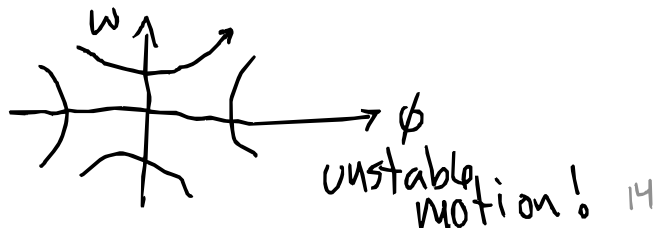
→ phase + energy oscillate out of phase.

For $\Omega^2 > 0$



For $\Omega^2 < 0$

hyperbolic solutions



For large amplitude oscillations

The eq'n of motion is modified:

$$\ddot{\phi} + \frac{\omega_s^2}{\cos \phi_s} (\sin \phi - \sin \phi_s) = 0$$

Note that: $\frac{d(\dot{\phi}^2)}{dt} = 2\dot{\phi} \ddot{\phi} = 2\ddot{\phi} \frac{d\phi}{dt}$

$$\begin{aligned} \rightarrow d\dot{\phi}^2 &= 2\ddot{\phi} d\phi \\ &= 2 \left(-\frac{\omega_s^2}{\cos \phi_s} (\sin \phi_s - \sin \phi) \right) d\phi \\ &= \frac{2\omega_s^2}{\cos \phi_s} \sin \phi d\phi - 2\omega_s^2 \sin \phi_s \overset{\text{tan}}{\frac{d\phi}{\cos \phi_s}} d\phi \end{aligned}$$

Integrate this:

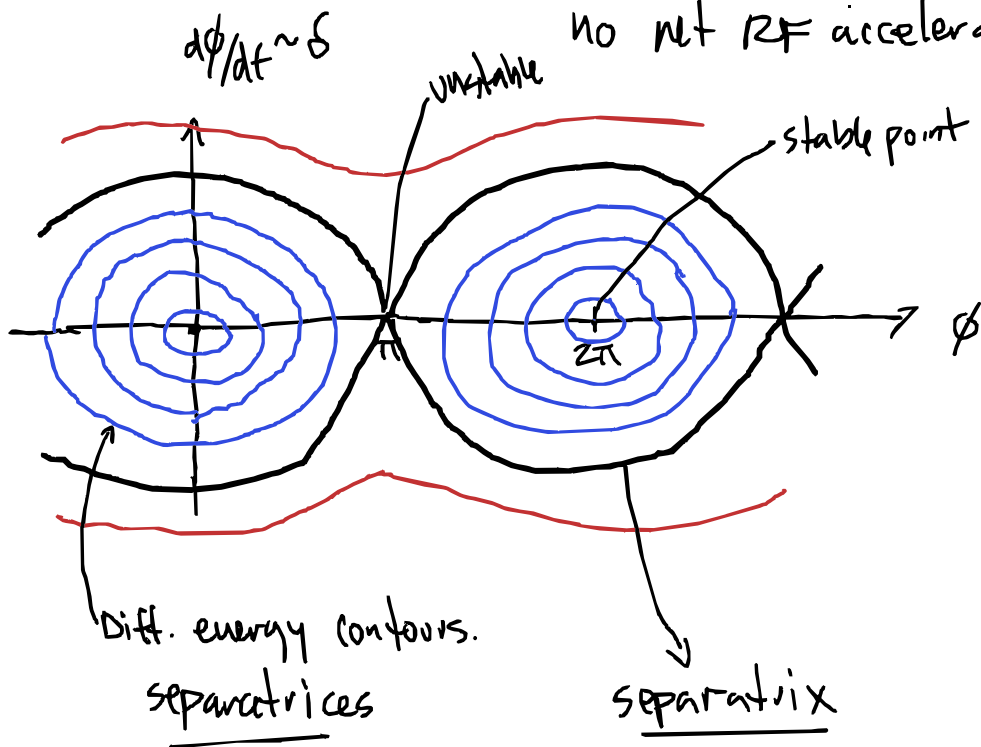
$$\frac{\dot{\phi}}{\omega_s} = \pm \sqrt{\frac{2(\cos \phi - \cos \phi_0)}{\cos \phi_s} + 2(\phi - \phi_0) \tan \phi_s + \frac{\dot{\phi}_0^2}{\omega_s^2}}$$

where ϕ_0 = phase at $t=0$

Let's look at Z-races:

① $\phi_s = 0 \rightarrow$ Normal pendulum motion

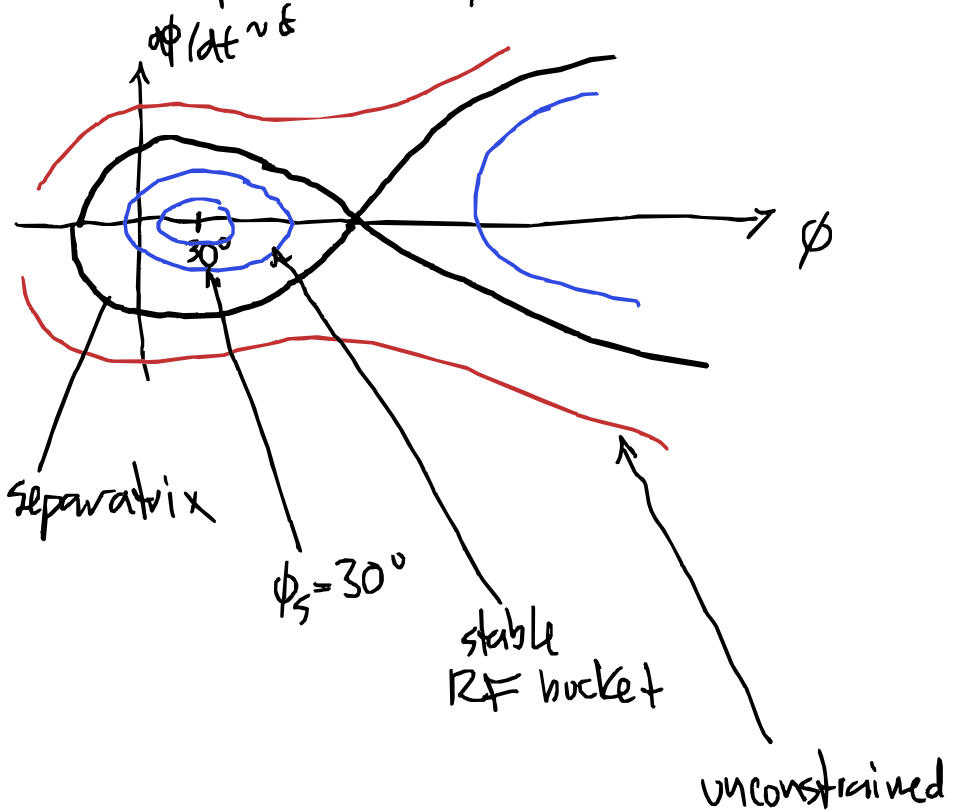
stable RF "buckets"
no net RF acceleration.



- Inside the separatrix $\rightarrow$ stable oscillations
RF bucket
- outside $\rightarrow$ particles are unconstrained
 $\Rightarrow$ No phase stability

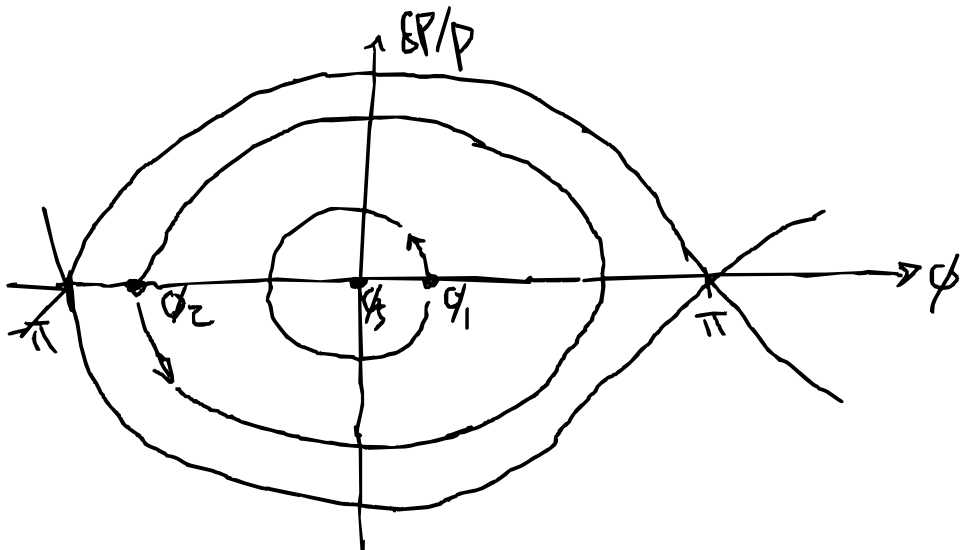
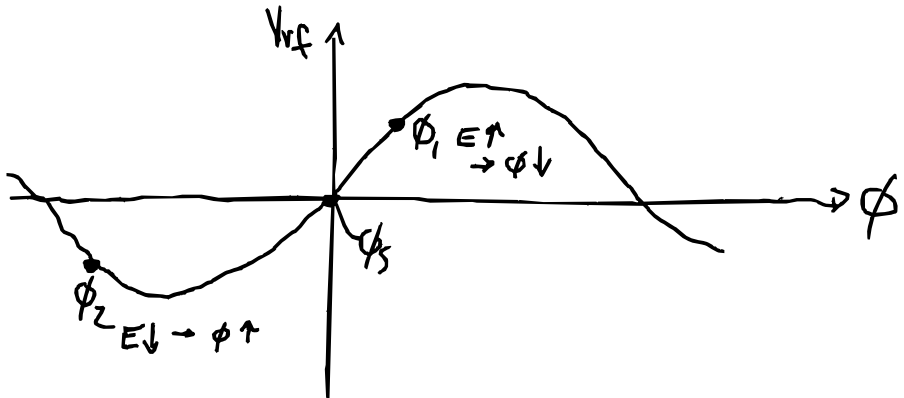
③ $\phi_s \neq 0 \rightarrow$ Acceleration!

the separatrices get skewed



How do particles move around the separatrix plot?

Case of $\psi_s = 0$



Hamiltonian Formalism

For small oscillations and $\phi_s = 0 \rightarrow$ no accel.

$$\text{eq. of motion} \rightarrow \ddot{\phi} + \omega^2 \phi = 0$$

For a simple mech. oscillator $\rightarrow \ddot{x} + \omega^2 x = 0$

$$\text{Hamiltonian} \rightarrow \mathcal{H} = \frac{1}{2} \frac{p^2}{m} + \frac{1}{2} m \omega^2 x^2$$

$$(\text{note } p \propto \frac{dx}{dt})$$

Without deriving it, the Hamiltonian for long. phase oscillations is:

$$\mathcal{H} = \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} \omega_s^2 \phi^2$$

For large oscillations:

$$\text{eq. of motion} \rightarrow \ddot{\phi} + \frac{\omega^2}{\cos \phi_s} (\sin \phi - \sin \phi_s) = 0$$

$$\rightarrow \mathcal{H} = \frac{1}{2} \dot{\phi}^2 - \omega^2 \cos \phi$$

What does this tell us?

The contours on separatrices are for fixed values of H .

- tells us:
- what phases are stable
 - what the RF bucket looks like.
 - how fast particles oscillate
 - momentum acceptance
- etc.

see Weidemann Ch. 9 for a more vigorous treatment of Hamiltonians!
